

**CROSSLINGUISTIC INFLUENCE
IN LANGUAGE AND COGNITION**

CROSSLINGUISTIC INFLUENCE IN LANGUAGE AND COGNITION

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Contents

<i>List of tables and figures</i>	ix
<i>Preface</i>	xi
1 OVERVIEW	1
1.1. Introduction	1
1.2. Phases of Transfer Research	4
1.3. Historical Skepticism about Transfer	8
1.4. Landmark Findings from the Pre-1990s Transfer Research	10
1.5. Recent Developments in CLI: New Areas of Research	13
1.6. Recent Developments in CLI: New Theoretical Accounts	15
1.7. Types of CLI	19
1.7.1. Area of Language Knowledge/Use	21
1.7.2. Directionality	21
1.7.3. Cognitive Level	22
1.7.4. Type of Knowledge	23
1.7.5. Intentionality	24
1.7.6. Mode	24
1.7.7. Channel	24
1.7.8. Form	25
1.7.9. Manifestation	25
1.7.10. Outcome	25
2 IDENTIFYING CROSSLINGUISTIC INFLUENCE	27
2.1. Introduction	27
2.2. Defining the Scope	28
2.2.1. Two Broad Approaches to Investigating CLI	28
2.2.2. Psycholinguistic Approaches to CLI	30

- 2.3. Intrasubjective Methods 32
- 2.4. Intersubjective Methods 36
 - 2.4.1. Pseudolongitudinal Designs: The Nexus Between Longitudinal and Cross-Sectional 36
 - 2.4.2. Evidence for CLI 41
 - 2.4.3. Methodological Rigor 48
- 2.5. Additional Methodological Considerations 52
 - 2.5.1. Mediating Variables 52
 - 2.5.2. Scope of Effects 58
 - 2.5.3. Promising Future Methods for Investigating Transfer 58

3 LINGUISTIC TRANSFER

61

- 3.1. Introduction 61
- 3.2. Phonological and Orthographic Transfer 62
 - 3.2.1. CLI in the Perception and Production of Segmental Phonology 63
 - 3.2.2. CLI Effects on Syllable Structure and Suprasegmental Phonology 67
 - 3.2.3. Orthographic Transfer 70
- 3.3. Lexical and Semantic Transfer 72
 - 3.3.1. Morphophonological versus Semantic Transfer in Language Users' Lexical Errors 75
 - 3.3.2. CLI Effects in Lexical Representation, Lexical Accessibility, and Lexical Activation 82
 - 3.3.3. Word Choice Transfer 88
- 3.4. Morphological and Syntactic Transfer 92
 - 3.4.1. Morphological Transfer 92
 - 3.4.2. Syntactic Transfer 96
- 3.5. Discursive, Pragmatic, and Sociolinguistic Transfer 102
 - 3.5.1. Discursive Transfer 102
 - 3.5.2. Pragmatic Transfer 106
 - 3.5.3. Sociolinguistic Transfer 109
- 3.6. Conclusions 111

4 CONCEPTUAL TRANSFER

112

- 4.1. Introduction 112
- 4.2. Language and Concepts 113
 - 4.2.1. Language-mediated Concepts 113

- 4.2.2. Conceptual Structure 116
- 4.2.3. Semantic and Conceptual Levels of Representation 118
- 4.2.4. Semantic and Conceptual Transfer 120
- 4.3. Crosslinguistic Differences and CLI in Eight Conceptual Domains 122
 - 4.3.1. Objects 123
 - 4.3.2. Emotions 125
 - 4.3.3. Personhood 129
 - 4.3.4. Gender 132
 - 4.3.5. Number 136
 - 4.3.6. Time 139
 - 4.3.7. Space 142
 - 4.3.8. Motion 145
- 4.4. Conclusions 148

5 CONCEPTUAL CHANGE

153

- 5.1. Introduction 153
- 5.2. Conceptual Change in Second Language Acquisition 154
- 5.3. Internalization of New Concepts 156
- 5.4. Restructuring 160
- 5.5. Convergence 164
- 5.6. Shift 166
- 5.7. Conceptual Attrition 169
- 5.8. Conclusions 171

6 TRANSFERABILITY AND FACTORS THAT INTERACT WITH TRANSFER

174

- 6.1. Introduction 174
- 6.2. Linguistic and Psycholinguistic Factors 176
 - 6.2.1. Crosslinguistic Similarity 176
 - 6.2.2. Area of Language Acquisition and Use 183
 - 6.2.3. Frequency, Recency, and Salience 183
 - 6.2.4. Markedness and Prototypicality 186
 - 6.2.5. Linguistic Context 189
- 6.3. Cognitive, Attentional, and Developmental Factors 190
 - 6.3.1. Level of Cognitive Maturity 190
 - 6.3.2. Developmental and Universal Processes of Language Acquisition 192

6.3.3. Cognitive Language Learning Abilities	193
6.3.4. Attention to and Awareness of Language	194
6.4. Factors Related to Cumulative Language Experience and Knowledge	197
6.4.1. Age	197
6.4.2. Length, Frequency, and Intensity of Language Exposure	199
6.4.3. Length of Residence	200
6.4.4. General Level of Proficiency	201
6.4.5. Number and Order of Acquired Languages	204
6.5. Factors Related to the Learning Environment	206
6.6. Factors Related to Language Use	207
6.7. Conclusions	210

7 CONCLUSIONS 211

7.1. Introduction	211
7.2. Recent CLI Research: Key Findings and Arguments	212
7.3. Implications for Applied Linguistics	216
7.4. Implications for Modeling of Bi- and Multilingual Representation, Processing, and Acquisition	219
7.4.1. Language Competence	219
7.4.2. Language Storage and Access	220
7.4.3. Interaction Between Languages	222
7.4.4. Semantic and Conceptual Levels of Representation	224
7.4.5. CLI in the Multilingual Lexicon	227
7.5. Implications for Future Research	229
7.5.1. Methodology	229
7.5.2. Research Areas	232

<i>References</i>	236
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<i>Index</i>	275
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Tables and Figures

TABLES

- 1.1 Phases of Transfer Research 5
- 1.2 Characterization of CLI Types Across Ten Dimensions 20
- 2.1 Hypothetical Data Representing a Stratified Design 38
- 2.2 Breakdown of Master's (1987) Learners by L2 Proficiency Level and L1 39
- 2.3 Rationale for Intragroup Homogeneity as Evidence of CLI 43
- 2.4 Rationale for Intergroup Heterogeneity as Evidence of CLI 45
- 2.5 Rationale for Crosslinguistic Performance Congruity as Evidence of CLI 46
- 2.6 Sample Group Structure: L1 Background as Independent Variable, L2 Proficiency as Mediating Variable 54
- 2.7 Sample Group Structure: L1 Background as Independent Variable, L2 Proficiency, Motivation, and Age as Mediating Variables 56
- 5.1 Typology of Conceptual Change Induced by L2 Learning 156

FIGURE

- 3.1 Three Levels of Lexical Representation 83

Preface

The study of crosslinguistic influence (CLI), or transfer, is peculiar among language acquisition and language use phenomena in at least two senses. First, empirical interest in CLI has existed since long before the formal establishment of the fields that now claim it as their own (cf., Larsen-Freeman & Long, 1991, p. 5; Odlin, 1989, p. 6). Second, unlike most other well-known factors affecting language acquisition and use (e.g., acculturation, anxiety, input, universal principles and parameters), which are often investigated from the perspective of a particular theoretical stance, research on transfer has tended to remain largely exploratory in nature—being driven mainly by theory-neutral research questions rather than by theory-specific hypotheses. This is not to say that transfer researchers have had no theoretical interest in CLI. Rather, it is probably because of the complicated nature, broad scope, and long history of interdisciplinary interest in the phenomenon that researchers have tended to follow a concatenate (or research-then-theory) approach (cf. R. Ellis, 1994, p. 474) to the study of CLI.

Each of the hundreds of empirical and theoretical studies on CLI has contributed to what we know about the phenomenon, but until now there has not been a major book-length review of the findings of such studies since Odlin (1989). The primary impetus for this book is our belief that it is once again time for a synthesis of the empirical and theoretical findings concerning CLI, especially with respect to the many important developments that have taken place since 1989. Additionally, although what the field now knows about CLI is undoubtedly dwarfed by what remains to be discovered, this line of inquiry has entered a new era since 1989. The new era exhibits four characteristics that distinguish it from earlier work on transfer. First, CLI is no longer a phenomenon for which there exist no principled criteria for confirming its presence or absence in language data (cf. Meisel, 1983). Second, CLI is no longer treated by researchers simply as a background, mediating, or intervening variable, examined only as an *explanans* (i.e., an explanation for the language behavior in question) and never as an *explanandum* (i.e., the phenomenon to be explained).

There were, of course, studies before 1989 that examined transfer as an explanandum (e.g., Andersen, 1983; Kellerman, 1978; Ringbom, 1978b; Selinker, 1969), but such studies were relatively sparse until the 1990s. Third, in the new era of transfer research, CLI is no longer a vague notion that lacks explicit theoretical status. Indeed, a major objective of this book is to explore, highlight, and detail its developing interdisciplinary theoretical status. Last but not least, inquiry into the nature of CLI now goes beyond the level of language knowledge and probes into its cognitive bases. Like the magnifying glasses portrayed on the cover of this book, languages differ with respect to the aspects of reality that they give prominence to. These differences have implications for how speakers of different languages perform both verbal and nonverbal tasks in either their native or a later-learned language, and they also have implications for how learning another language can lead to changes in a person's conceptual knowledge. Against this backdrop, a second major objective of this book is to demonstrate how the scope of CLI extends beyond linguistic transfer into areas that we will describe as conceptual transfer.

The present book consists of seven chapters. Chapter 1 provides a summary of historical landmarks in the investigation of transfer, a discussion of important new directions in CLI research, and a scheme for understanding the relationship between different types of CLI. Chapter 2 discusses methodological issues and approaches for identifying and measuring the effects of CLI. Chapter 3 reviews recent studies on linguistic transfer in several different areas of language knowledge and use, including, among others, phonology, lexis, semantics, morphology, syntax, discourse, and pragmatics. In Chapter 4, we discuss findings and issues related to conceptual transfer in several domains of reference, including, among others, gender, number, space, and motion. In Chapter 5, we discuss the interaction between the languages that a person knows from a longitudinal perspective, especially in relation to how experience in more than one language leads to changes in one's conceptual system and in the associations that conceptual representations share with linguistic representations. In Chapter 6, we discuss the variables that both promote and constrain instances of transfer, such as language proficiency, perceived distance between the source and recipient languages, recency and activation levels of the languages in question, the L2 status effect, and several other factors. Finally, in Chapter 7, we describe a number of implications that the recent transfer literature holds for theory, research, and pedagogy.

We conclude our preface by acknowledging and thanking those people who have been most influential in our writing of this book. We are especially thankful to Vivian Cook, Terence Odlin, and Håkan Ringbom for their helpful suggestions and insightful comments on a preliminary version of this book. We also thank our editor, Cathleen Petree, for her encouragement, help, and especially her patience during the years that it took us to complete our writing and revising of

this book. Finally, we wish to thank our family members, friends, and colleagues for their support and encouragement on this project, and for providing us with a wealth of information, insights, and examples of crosslinguistic influence in both language and cognition.

CHAPTER 1

Overview

1.1. INTRODUCTION

Crosslinguistic influence—or the influence of a person’s knowledge of one language on that person’s knowledge or use of another language—is a phenomenon that has been of interest to laypeople and scholars alike since antiquity and most likely ever since language evolved. One of the earliest references to language contact, bilingualism, and crosslinguistic influence comes from Homer’s *Odyssey*, where Odysseus tells Penelope about the “mixed languages” of Crete. Due to widespread multilingualism in the ancient world, instances of crosslinguistic influence abound in a variety of ancient texts, ranging from epitaphs and personal letters, to legal and commercial documents, to religious and literary treatises (Adams, Janse, & Swain, 2002). These texts also offer evidence of negative attitudes towards the phenomenon of *transfer* (another term for crosslinguistic influence), such as derogatory remarks about “speakers of bad Greek” made by ancient writers and philosophers, including Homer, Herodotus, and Flavius Philostratus. In fact, Janse (2002) argues that the negative term *barbarians* and its derivatives were commonly used to refer not only to speakers of languages other than Greek but also to foreigners speaking “bad Greek” or, in our contemporary terms, foreigners exhibiting first language transfer.

The trend of seeing language transfer as a negative phenomenon, associated with low moral character and limited mental abilities, persisted all the way into the twentieth century when increased global migration fueled the fear of foreigners and of the unspeakable things they could do to one’s language. In a speech delivered in 1905 to the graduating class of Bryn Mawr College, Henry James, traumatized by his recent visit to Ellis Island and to the Lower East Side, populated by recently arrived Eastern European immigrants, warned the students against an imminent threat to the civilized tongue from

the vast continent of aliens whom we welcome, and whose main contention . . . is that, from the moment of their arrival, they have just as much property

in our speech as we have, and just as good a right to do what they choose with it. (cited in Brumberg, 1986, pp. 6–7)

In reality, it was the immigrants' native language that was in danger of second language influence. This influence is well-documented in Mencken's (1937) monumental treatise *The American Language*. Mencken's discussion of the influence of English on the immigrants' native languages offers a glimpse of the interest that lexical (word-related) and syntactic (grammar-related) transfer from the second language to the first elicited in the scholarly community. At the same time, many scholars and writers of the era, including some well-educated immigrants, frowned on both first and second language transfer. Marcus Ravage, a well-known writer and himself a Romanian immigrant, criticized the influence of Yiddish and Romanian on his fellow Romanians' English:

My friends were finding English contemptibly easy. That notion of theirs that it was a mixture of Yiddish and Rumanian, although partly justified, was yielding some astonishing results. Little Rumania was in throes of evolving a new tongue – a crazy-quilt whose prevailing patches were, sure enough, Yiddish and Rumanian, with here and there a sprinkling of denatured English. They felt no compunction against pulling up an ancient idiom by the roots and transplanting it bodily into the new soil. One heard such phrases as "I am going on a marriage," "I should live so," "a milky dinner." (Ravage, 1917, p. 103)

Ravage's concerns were echoed by another well-known immigrant writer of the era, Abraham Cahan, who poked fun at the lexical and syntactic influence of English on the immigrants' Yiddish:

I have already described how the Yiddish of American-born children grated on my ears. The Americanized Yiddish of the immigrants, studded with English expressions, was no better. My anger rose when I heard such expressions as "er macht a leben" (he makes a living) or "er is vert tsehn toisend dolar" (he is worth ten thousand dollars). Or such horrors as "vindes" (windows) or "silings" (ceilings) and "pehtaytess" (potatoes). (Cahan, 1926, pp. 241–242)

These examples serve to show that, in the absence of an in-depth understanding of the workings of language, transfer phenomena often came to signify sloppiness, narrow-mindedness, and lack of mental clarity and sound thinking. Linguists and psychologists often contributed to this picture, arguing, for instance, that the mutual interference of languages represents a danger to sound thinking (Epstein, 1915) and that first language transfer in pronunciation is due to the learners' laziness and lack of interest in changing their phonological behavior (Jespersen, 1922). These convictions were not truly challenged until the

1940s and 1950s, when the work of Charles Fries (1945), Uriel Weinreich (1953), Einar Haugen (1953), and Robert Lado (1957) moved discussions of language transfer to a scholarly footing, legitimizing transfer as an unavoidable feature of language learning and use and exploring it as a linguistic, psycholinguistic, and sociolinguistic phenomenon. Weinreich's work in particular was groundbreaking in terms of both the breadth and depth of the empirical evidence he examined, and in terms of the insights he offered about the nature of CLI. His book *Languages in Contact* (1953) is still frequently cited in transfer research.

Since the 1950s, a number of additional books have dealt extensively with transfer, including, in chronological order, Vildomec (1963), Gass and Selinker (1983), Kellerman and Sharwood Smith (1986), Ringbom (1987), Dechert and Raupach (1989), Odlin (1989), Gass and Selinker (1992), Sjöholm (1995), Jarvis (1998), Kecskes and Papp, Cenoz, Hufeisen, and Jessner (2001, 2003), Alonso (2002), Cenoz, Hufeisen, and Jessner (2003), Cook (2003), Arabski (2006), and Ringbom (2007). Of these, Odlin's book stands out as having provided the broadest synthesis of the transfer literature to date. The present book is not intended as a replacement for Odlin's. Rather, our aim is, first, to characterize the new developments that have taken place in transfer research since the publication of his book and, second, to link these developments to ongoing interdisciplinary inquiry into language and cognition. The scope of our book is also somewhat narrower than Odlin's in that we deal with transfer almost exclusively in relation to adult second language users, and almost exclusively as a psycholinguistic phenomenon—or as a phenomenon that takes place in the minds of individuals, and which is subject to the effects of various cognitive, linguistic, social, and situational factors. In this book, therefore, we do not delve into issues related to child bilingualism, transfer as a societal phenomenon, or code-switching.

We use the terms *transfer* and *crosslinguistic influence* interchangeably as theory-neutral cover terms to refer to the phenomenon in question, even though we recognize that by the 1980s some researchers no longer considered the term “transfer” to be a suitable label for the phenomenon because of its traditional association with the behaviorist notion of skills transfer (see Lado, 1957, p. 2; Odlin, 1989, p. 26; Osgood, 1953, p. 520; Selinker, 1983, p. 34). *Interference* is another term that has often been used since Weinreich (1953), but this term also carries behaviorist connotations and additionally has the disadvantage of directing one's attention only to the negative outcomes of transfer. In the mid 1980s, Kellerman and Sharwood Smith (1986) proposed the term *crosslinguistic influence* as a theory-neutral term that is appropriate for referring to the full range of ways in which a person's knowledge of one language can affect that person's knowledge and use of another language. This term has since gained general acceptance in the fields that investigate this phenomenon, even though the terms “transfer” and “interference” have continued to be used synonymously with it. More recently, some scholars have suggested that even “crosslinguistic influence”

may be an inappropriate term to refer to the phenomenon, given that the influence of one language on another in an individual's mind may be more an outcome of an integrated multicompetence than of the existence of two (or more) completely separate language competences in the mind (e.g., Cook, 2002). We discuss this perspective more fully in section 1.6 of this chapter, and in the meantime simply state that although the suitability of the terms "transfer" and "crosslinguistic influence" can certainly be called into question, these are at present the most conventional cover terms for referring to the phenomenon, and are the terms that we will use throughout this book. The term *first language* (L1) will be used to refer to the first language acquired by the speaker from a chronological perspective, even if this language is no longer the speaker's dominant language. The term *second language* (L2) will refer to any language acquired subsequently, regardless of the context of acquisition or attained level of proficiency.

The two primary aims of the present chapter are to provide a brief historical perspective of the important developments that took place in the transfer research prior to the 1990s (or prior to the 1989 publication of Odlin's book), and to offer a summary of the new developments that have taken place since the beginning of the 1990s. We begin with the historical perspective in section 1.2 by describing the four general phases through which the investigation of a psycholinguistic phenomenon logically progresses, and by discussing which of these phases transfer research has already passed through, and where it now stands. In section 1.3, we discuss some of the historical skepticism about the scope and importance of transfer, and explain why much of the skepticism was unwarranted and why it has largely given way to a general recognition of both the importance and the complexity of the phenomenon. Section 1.4 concludes our description of the historical perspective by outlining what we consider to be the primary landmark findings of the pre-1990s transfer research. In section 1.5, we turn to a summary of the important developments that have taken place in crosslinguistic influence (CLI) research since the beginning of the 1990s, giving special attention to new areas of empirical investigation. Section 1.6 continues our summary of recent developments by focusing on new theoretical accounts of CLI that have arisen since the beginning of the 1990s. Finally, in section 1.7, we draw together the historical and recent perspectives by clarifying the relationship among the different types of CLI that have been referred to throughout the literature.

1.2. PHASES OF TRANSFER RESEARCH

Transfer research, like the investigation of various other phenomena in language and cognition, can be described as progressing through four general phases. During Phase 1, the phenomenon in question gains recognition as a possible

explanans (explanation, affecting factor, or independent variable) for what is considered to be a more important *explanandum* (thing to be explained, or dependent variable). Next, during Phase 2, the phenomenon in question comes to be considered important enough to be investigated as an explanandum in its own right, with its own set of empirical *explanantia* (or factors that affect its behavior). Then, in Phase 3, the phenomenon attracts sufficient theoretical interest to give rise to sophisticated competing theoretical models and hypotheses concerning the social, situational, and mental constraints, constructs, and processes involved in the occurrence of the phenomenon; empirical research during this phase is thus highly theory-driven. Finally, Phase 4 is characterized by a complex understanding of the phenomenon in terms of the actual neurophysiological structures and processes through which it operates. These phases and their primary characteristics are listed in Table 1.1.

TABLE 1.1
Phases of Transfer Research

General Description		Primary Research Concerns
Phase 1	Recognition and investigation of the phenomenon as a factor—as an explanans or intervening or independent variable—that affects other processes (such as second language acquisition)	• Identifying cases of transfer • Defining the scope of transfer • Quantifying transfer effects
Phase 2	Investigation of the phenomenon as a primary process itself—as an explanandum or dependent variable—that has its own set of explanantia or independent variables	• Verification of transfer effects • Identifying causes of transfer • Identifying constraints on transfer • Investigating the selectivity of transfer • Investigating the directionality of transfer effects
Phase 3	Development of theories designed to explain the phenomenon in relation to social, situational, and mental constraints, constructs, and processes	• Development of theoretical models that explain how, why, when, and what types of CLI occur • Development of specific, testable hypotheses concerning CLI • Empirical testing of these hypotheses

TABLE 1.1 *continued*
Phases of Transfer Research

<i>General Description</i>	<i>Primary Research Concerns</i>
Phase 4 Development of a precise physiological account of how the phenomenon takes place in the human brain	• Detailed mapping of the brain in relation to how language is acquired, stored, and processed • Accumulation of direct evidence of crosslinguistic neurological connections in a person's long-term memory—of how such connections are formed, changed, and maintained • Accumulation of direct evidence of how languages are activated in the brain and of how a person's knowledge of one language can be activated and interfere with his or her use of another language

From our perspective, Phase 1 of transfer research began when transfer became widely recognized as a variable that can affect language acquisition, language use, and other linguistic, psychological, cognitive, and cultural processes. Its beginning was marked by the recognition of transfer as an intervening variable in language acquisition and language change, and this was followed by studies that examined transfer in a more controlled way as a moderator or primary independent variable (for a description of different types of variables, see, e.g., Hatch & Lazaraton, 1991, pp. 63–68). It is difficult to put an exact date on the beginning or end of the first phase. It may have begun informally centuries ago, but its beginning was clearly underway in the mid to late 1800s with studies by Müller (1861) and Whitney (1881), and a little later by Epstein (1915) and others (see, e.g., Odlin, 1989, pp. 6–9). Perhaps the most substantial contribution to Phase 1 occurred in the 1950s in the form of Weinreich's (1953) detailed examination of numerous types of transfer (which he called interference), and his discussion of methods for identifying and quantifying transfer and its relationship to other aspects of bilingualism. Phase 1 continued at least until the mid 1970s, and remnants of this phase can still be seen in some of the transfer research today. The issues that were fundamental to researchers during the peak

of Phase 1 were (a) the identification of transfer, especially in relation to which learner errors are due to L1 influence (e.g., Dušková, 1969); (b) defining the scope of transfer, especially in relation to the areas of language use are affected by it (e.g., Weinreich, 1953); and (c) the quantification of transfer effects, especially in relation to the proportion of errors it accounts for in comparison with other variables (e.g., Grauberg, 1971).

As mentioned, there is no exact date for the end of Phase 1 of transfer research; all three of these issues are still being pursued in empirical studies today. Nevertheless, Weinreich laid the groundwork for the following phase in the 1950s, and a shifting trend out of Phase 1 can clearly be seen in work published in the 1960s by Selinker (1969) and others (see, e.g., Selinker, 1992); this shift to Phase 2 was probably fully underway by the mid to late 1970s with studies by Kellerman (1978), Ringbom (1978b), Schachter (1974) and others. During Phase 2 of transfer research, the identification, scope, and quantification of transfer effects have remained important issues, but to these have been added the following important issues: (a) verification of transfer effects (e.g., Ard & Homburg, 1983; Jarvis, 2000a; Selinker, 1969), (b) sources and causes of transfer (e.g., Cenoz, 2001; Ringbom, 1978b), (c) constraints on transfer (e.g., Andersen, 1983; Kellerman, 1978), (d) the selectivity of transfer at the level of individual learners (e.g., Giacobbe & Cammarota, 1986; Schmidt, 1977; Zobl, 1980), and (e) the directionality of transfer effects (e.g., Ringbom, 1987; Kecskes & Papp, 2000). All of these issues are still the focus of a great deal of attention.

Phase 2 probably has not ended yet, but a shift to the next phase has nevertheless already begun. Phase 3 is a phase of increasing theoretical interest in CLI, a phase where there is keen interest in modeling, explaining, and empirically investigating the mental constructs and processes through which CLI operates. Elements of Phase 3 were certainly evident already in the work of Lado (1957), Weinreich (1953), and Vildomec (1963), and later in work related to, for example, the UG paradigm (e.g., White, 1989) and the Competition Model (e.g., Harrington, 1987), but it was not until fairly recently that we began to see theoretical models and frameworks designed specifically to explain CLI and the interaction between languages in the mind (e.g., Cook, 1991, 2002; Flege, 1995; Kroll & De Groot, 1997; MacWhinney, 2005). We will discuss some of these proposals in section 1.6 of this chapter and will return to them in the conclusions chapter of the book.

While the shift from Phase 2 to Phase 3 is probably still underway, some of the necessary groundwork for the highly anticipated Phase 4 is already being laid. The groundwork for Phase 4 can be seen in studies that probe into the neurophysiology of how languages are stored and how they operate in the brain, especially in relation to the brains of people who know and use more than one language. Neurolinguistic research relying on positron emission tomography (PET), functional magnetic resonance imaging (fMRI), electroencephalographic

and magnetoencephalographic recording (EEG), magnetic source imaging and event-related brain potentials (ERPs), and the analysis of aphasic symptoms and recovery has already brought to light a number of important findings concerning the interaction between languages within the brain (see, e.g., Emmorey et al., 2005; Fabbro, 1999, 2002; Franceschini, Zappatore & Nitsch, 2003; Green, 2001; Hahne, 2001; Hahne & Friederici, 2001; Hasegawa, Carpenter & Just, 2002; Nakada, Fujii & Kwee 2001; Paradis, 2004; Schumann, 2001). However, Phase 4 probably will not be solidly underway for several more decades—not until the tools of neurolinguistic research allow for a more precise description of the exact nature of individual phonemic, morphological, lexical, semantic, conceptual, syntactic, discursive, pragmatic, and collocational representations in the brain, and not until we understand the precise neurolinguistic and neurophysiological processes through which these representations are acquired, stored, and accessed, and how the representations of two different languages might be associated or even integrated. It is also quite possible that at least some of these phenomena will never be captured on the neurolinguistic level.

In summary, there is quite a bit of overlap among the phases of transfer research. Once a phase is begun, it may never completely end, given that the findings and concerns from earlier phases remain relevant after subsequent phases have begun. On the other hand, assumptions about the nature of language espoused at the earlier stages may be abandoned or transformed at subsequent stages. At present, CLI research appears to be in a transition from Phase 2 to Phase 3, or from concerns about the measurement of transfer and the factors that interact with it to more full-fledged theoretical explanations of how CLI operates. In a word, we are at a point where we have amassed a truly extensive amount of knowledge about the phenomenon, but we also know that we have a long way to go before we have explained it fully.

1.3. HISTORICAL SKEPTICISM ABOUT TRANSFER

Even though the fields of research that investigate transfer still have a long way to go in their understanding of this phenomenon, we can already see that CLI is both fascinating and multifaceted—a phenomenon that has the potential to percolate into all areas of language knowledge and use among those who know more than one language. Its complexity has not always been acknowledged, however, and even recently, in our interactions with other scholars we have heard some maintain inexorably that transfer is nothing more than falling back on a language that one already knows when lacking knowledge in the language that one is presently learning. This claim, known as the *ignorance hypothesis*, was originally introduced by Newmark (1966), and has been restated by Newmark and Reibel (1968), Krashen (1983), and a few others. The essence

of this hypothesis can be seen in the following quotation from Newmark and Reibel:

a person knows how to speak one language, say his native one; but in the early stages of learning his new one, there are many things that he has not yet learned to do. . . . What can he do other than use what he already knows to make up for what he does not know? To an observer who knows the target language, the learner will be seen to be stubbornly substituting the native habits for target habits. But from the learner's point of view, all he is doing is the best he can: to fill in his gaps of training he refers for help to what he already knows. (1968, p. 159)

A corollary of this view of language transfer is the denial of the existence of L2 influence on the L1. Take, for instance, an off-hand comment made recently by Adams and Swain (2002) about the impossibility of transfer from L2 Greek into L1 Latin in ancient public inscriptions found in Greece and made presumably by L1 Latin speakers: "*Unintentional* interference (from the writers' L2!) is out of the question" (p. 5). While we do not dispute the authors' contention about the origins of particular syntactic constructions, what is interesting to us is their use of parentheses and the exclamation point, which implicitly suggest that L2 influence on the L1 (in the L2 context to boot) is so highly unlikely that to posit it would be ridiculous.

Returning to the ignorance hypothesis, no one who is familiar with second language acquisition (SLA) and bilingualism/multilingualism would of course deny that CLI is sometimes triggered by a person's ignorance of a form, structure, or rule in one of the languages that he or she otherwise knows. Resorting in such cases to one's knowledge of another language is, in our experience, a fairly common communicative strategy (cf. Dörnyei, 1995). Even so, a crucial fact about CLI that this hypothesis ignores is that transfer occurs not only as a communicative strategy, but also very frequently as a learning strategy by which the learner uses his or her knowledge of one language as a resource for formulating hypotheses about the forms, structures, functions, meanings, rules, and patterns of another (e.g., R. Ellis, 1994, p. 314). In many cases, this results in hybrid structures that combine elements of both languages, such as when a Swedish speaker writing in English creates the word *luckly* to mean happy, based on her combined knowledge of Swedish (*lycklig* = happy) and English (*lucky*) (cf. Ringbom, 1987, p. 154). In other cases, CLI as a learning strategy involves associating forms (e.g., words or phrases) in one language with meanings acquired through another language, such as when Finnish speakers overgeneralize the use of the English preposition *in* to denote an internal goal (e.g., "They escaped in the police car" meaning "They escaped from the police car")—a pattern that is motivated by the nature of the Finnish spatial system (in which the corresponding form carries the

meaning of “from inside”) and which differs from the way speakers of other L1s overgeneralize the use of *in* (Jarvis & Odlin, 2000). Additionally, as Kleinmann (1977) showed, even when learners know the same structures, CLI can affect which particular structures they choose to use in a given context. The final blow to the ignorance hypothesis is dealt by the fact that CLI occurs not just from L1 to L2, but also from L2 to L1, even in cases where it is clear that *the L2 user* (i.e., someone who knows and uses more than one language) has not lost or forgotten the L1, not even the particular L1 structure that exhibits L2 influence (e.g., Jarvis, 2003). In all of these cases, it is clear that CLI is not simply a matter of falling back on one language when a person lacks pertinent knowledge of another language.

Some of today’s lingering adherents to the ignorance hypothesis were once among the cadre of enlightened researchers in the 1960s and 1970s who recognized that second language acquisition is a far richer and more complex phenomenon than simply the struggle to overcome L1 habits. Many of that enlightened corps, however, strayed into adopting and perpetuating the theory-induced but logically and empirically unsound notion that transfer is uninteresting or even negligible (e.g., Dulay & Burt, 1974; George, 1972; Krashen, 1981; Whitman & Jackson, 1972). This notion was fueled partially by the demise of behaviorism, and perhaps even more by the failure of contrastive analysis to predict learner errors with sufficient accuracy, as well as by the findings of a few studies that showed that transfer-related errors constituted only a small percentage of the total observed errors (for more discussion of the problems of these early studies, see, e.g., R. Ellis, 1994; Gass & Selinker, 2001; Odlin, 1989). However compelling those reasons seemed at the time, with hindsight it is tempting to draw an analogy between the transfer skeptics and hypothetical meteorologists who consider precipitation to be uninteresting because their instruments do not predict it accurately enough, or between the transfer skeptics and hypothetical meteorologists working in Southern California who generalize from the arid conditions in which they conduct their research that precipitation is negligible around the globe.

1.4. LANDMARK FINDINGS FROM THE PRE-1990s TRANSFER RESEARCH

Fortunately, the notion that transfer is uninteresting, negligible, and/or the result of ignorance or sloppy thinking is gradually becoming only a smudge in the history of the field. For this fact, we can thank those many researchers who throughout the 1960s–1980s never lost sight of the important and broad-reaching (though often subtle) effects of CLI, and who have established much of what we now know about the phenomenon. Following are eight landmark findings

concerning CLI that they brought to light during the 1960s–1980s. (These and other findings from the pre-1990s transfer research are covered more fully in Odlin [1989].)

1. Errors are not the only outcome of CLI. The consequences of CLI in many cases are positive, such as when it leads to conventional language use and accelerated language acquisition (e.g., Ard & Homburg, 1983; Ringbom, 1978a). Additionally, the similarities and differences between the *source language* (a language the language user knows that serves as the source of CLI effects) and the *recipient language* (the language where CLI effects occur) often lead not to errors per se, but to the general underproduction (of, e.g., relative clauses; Schachter, 1974) or overproduction (of, e.g., sentences beginning with *It is* and *There is*; Schachter & Rutherford, 1979) of structures in the recipient language. Moreover, CLI often manifests itself in the form of preferences for using certain language structures instead of others (e.g., one-part verbs instead of phrasal verbs—such as *enter* versus *come in*; Dagut & Laufer, 1985), where more than one preference is fully acceptable in the recipient language.
2. CLI can affect not only the rate and ultimate success of learners' second language acquisition; it can also affect the route of their acquisition—or the stages and sequences they pass through as they acquire proficiency in the target language (e.g., Schumann, 1979; Stauble, 1984). For example, it can affect the order in which Chinese and Spanish speakers acquire *this* versus *the* in L2 English (Zobl, 1982).
3. Contrary to the strong version of the Contrastive Analysis Hypothesis (e.g., Lado, 1957; see Odlin, 2006; Wardhaugh, 1970), differences between the source and recipient languages do not necessarily lead to learning difficulties or to CLI. Instead, easily perceivable differences often make target-language structures easier to acquire. This appears to be the case with the English progressive, which does not have a counterpart in Arabic but is nevertheless relatively easy for Arabic speakers to acquire (Kleinmann, 1977). Additionally, it is similarities (such as similar words and word-order rules) instead of differences between the source and recipient languages that most often appear to lead learners to make mental associations or *interlingual identifications* (see Odlin, 1989, pp. 113–114; Weinreich, 1953, p. 7) between structures in the two languages (e.g., Andersen, 1983).
4. CLI does not decrease linearly as competence and proficiency in the recipient language increase. Some transfer errors (such as Spanish speakers' use of an infinitive after a modal—e.g., *She can't to understand*; Taylor, 1975, p. 104) do tend to decrease with increases in recipient-language

proficiency, but in many other cases CLI shows up only at later stages of development when the learner has acquired enough of the recipient language to recognize the similarities between it and the source language. For example, Wode (1977) found that German-speaking learners of English do not show influence from German in the earliest stage of their acquisition of English negation (where they produce negations such as *No play baseball*), but they do show influence from German at a later stage (e.g., *I'm steal not the base*).

5. Language transfer can occur not only from an L1 to an L2 (*forward transfer*), but also from an L2 to an L3 (*lateral transfer*) (e.g., Ringbom, 1978b), and from an L2 to an L1 (*reverse transfer*) (e.g., Jakobovits, 1970; see also Weinreich, 1953). This has been a somewhat forgotten or ignored finding, but has received renewed interest in recent years, as we describe in section 1.5.
6. CLI interacts with other factors that together determine the likelihood of transfer—or the *transferability*—of a given structure in a given context. Some of the factors that affect transferability are language users' age (e.g., Zobl, 1983), their perception of the language distance (or typological relatedness) between the languages they know (e.g., Ringbom, 1978b), the degree to which language users perceive the language structures in question to be language-specific (e.g., learners' judgments about the "coreness" of various meanings of the verb *to break*; Kellerman, 1978), and the degree to which CLI coalesces with developmental sequences and universal learning principles (e.g., Andersen, 1983; Gass, 1983; Schumann, 1979; Zobl, 1980).
7. Transfer effects are not limited to language forms, such as the phonological or morphological structure of a word or the syntactic structure of a clause. Transfer effects also extend to the meanings and functions that language users associate with those forms, as can be seen, for example, in differences between English speakers and Spanish-speaking learners of English regarding what they classify as shoes versus boots (Graham & Belnap, 1986). Transfer effects also extend beyond forms and meanings to include the ways in which language is used to perform pragmatic functions, such as the ways in which Hebrew speakers from various L1 backgrounds perform an apology (e.g., Olshtain, 1983), and the ways in which Japanese-speaking learners of English refuse an offer (e.g., Takahashi & Beebe, 1987).
8. Individual differences between language users can lead to individual differences in the types and extent of CLI they exhibit in their use of the recipient language. For example, Odlin (1989) refers to studies that suggest that factors such as anxiety and aptitude can result in individual differences in learners' language performances, including CLI-related

differences pertaining to the degree to which they exhibit an L1 accent and the degree to which they avoid certain structures in the target language (pp. 130–132).

For most scholars, these landmark findings have laid to rest the idea that CLI is either uninteresting or negligible. More specifically, the empirical record has shown that transfer cannot be characterized simply as habit-related performance, as falling back on the L1, or as a mechanical process of linguistic transference from one language to another. Instead, the landmark findings, especially findings 3 to 8 above, have underscored that CLI is a highly complex cognitive phenomenon that is often affected by language users' perceptions, conceptualizations, mental associations, and individual choices.

1.5. RECENT DEVELOPMENTS IN CLI: NEW AREAS OF RESEARCH

More recent developments in transfer research have expanded on these classical findings, and have led to the recognition of more types of transfer, more constraints on transfer, and more ways in which the languages that a person knows can interact with one another. The new developments involve (a) new findings and refinements to already existing areas of research, (b) entirely new areas of research, and (c) new theoretical accounts of CLI. All three types of new developments are given attention throughout this book, but (b) and (c) deserve special attention because they represent not just new developments, but in fact new directions in CLI research. We summarize the new directions related to (b) in this section, and turn to a summary of (c) in the following section.

A perusal of Odlin's book reveals that by 1989 CLI research had flourished in the study of word order, relativization, negation, vocabulary, segmental phonology, and speech acts. However, relatively little research had been conducted on CLI in discourse, in particular with regard to narrative structure or suprasegmental phonology, although these areas were clearly marked as important by Odlin. Even less attention had been paid to transfer effects in perception and comprehension. In the years that have elapsed since his book was first published, CLI research has substantially expanded its scope in terms of topics, directionality, the number of languages considered, and areas of language use and processes where it has been explored.

Topic-wise, explorations of morphosyntax have begun to consider CLI in the use of phrasal verbs (e.g., Sjöholm, 1995), causative constructions (e.g., Helms-Park, 2001), gender assignment (e.g., Dewaele & Veronique, 2001) and linguistic framing (e.g., Pavlenko, 2003a; Pavlenko & Driagina, 2007; Pavlenko & Jarvis, 2002; Slobin, 1996; Vermeulen & Kellerman, 1999). Studies in phonology have

begun to examine suprasegmental transfer (e.g., Andrews, 1999; Gibson, 1997; Leather & James, 1996), and studies in discourse have begun exploring CLI in narrative construction (e.g., Berman, 1999; Kellerman, 2001; Pavlenko, 2002b, 2003b; Verhoeven & Strömquist, 2001) and in discourse styles and conversational strategies, such as back-channeling (e.g., Heinz, 2003; Tao & Thompson, 1991).

With regard to directionality, CLI effects have been examined in terms of forward transfer (e.g., Jarvis, 2000a), reverse transfer (e.g., Andrews, 1999; Cook, 2003; Kecskes & Papp, 2000; Major, 1992; Pavlenko, 2000), lateral transfer (e.g., Cenoz et al., 2001), and bidirectional transfer (e.g., Pavlenko & Jarvis, 2002; Su, 2001). The investigation of reverse transfer, or L2 (L3 etc.) influence on L1, is a particularly noteworthy development in this area. As will be shown in the next section, the acknowledgment of this type of transfer has important theoretical implications for our understanding of linguistic competence and native-speakerness.

Another important and long-awaited development is the expansion of the scope of CLI research to include the interaction between three or more languages in the mental lexicon (e.g., Dewaele, 1998; Cenoz et al., 2001, 2003), which shows that later learned languages may affect one another as well as the L1, and that this interaction may facilitate further learning and engage the learners' creativity. Topics such as these are now frequently discussed at the International Conference on Third Language Acquisition and Multilingualism, which has been held every second year since 1999.

The final important development is the expansion of areas of language use in which transfer has been explored. Traditionally, transfer was examined in terms of particular linguistic subsystems, such as syntax and semantics, in the areas of second language production and acquisition (the latter as seen in production). Recent studies have pushed the boundaries of the field outward, demonstrating that CLI is not limited to production and acquisition and can be identified in a variety of psycholinguistic processes. To date, CLI has been examined in lexical and syntactic processing (e.g., Cook et al., 2003; Costa et al., 2003a; Dijkstra, 2003; Dijkstra, Van Jaarsveld & Ten Brinke, 1998; Kroll & Dussias, 2004; Von Studnitz & Green, 2002), in listening and reading comprehension and sentence interpretation (e.g., Ringbom, 1992; Su, 2001; Upton & Lee-Thompson, 2001), in tip-of-the-tongue states (e.g., Ecke, 2001; Gollan & Silverberg, 2001), in nonverbal communication (e.g., Brown & Gullberg, 2005; Kellerman, 2001; Kellerman & Van Hoof, 2003; McCafferty, 2002, 2004; Negueruela et al., 2004), and in conceptual representation (e.g., Andrews, 1994; Jarvis, 1998; Kroll & De Groot, 1997; Pavlenko, 1997, 2003a, 2005a). In the area of conceptual representation, intriguing transfer effects have been detected in the interpretation of and reference to emotions, which were previously considered to be universal rather than language-specific (e.g., Panayiotou, 2004a, b; Pavlenko, 2002b, 2005b; Pavlenko & Driagina, 2007; Stepanova Sachs & Coley, 2006).

Considerable advances have also been made in the study of second language writing (Cook & Bassetti, 2005a), in particular from the perspective of contrastive rhetoric (e.g., Connor, 1996; Kecskes & Papp, 2000). Lately, interest in and attention to transfer effects have been exhibited by classicists who have explored bilingualism and language contact in antiquity, through the study of bilingual and multilingual texts and translations (e.g., Adams, 2003; Adams, Janse, & Swain, 2002). Based on the findings of the research on writing and literacy, be it ancient or contemporary, several scholars have begun to argue that biliteracy is a phenomenon distinct from bilingualism, and thus transfer effects pertinent to biliteracy should be studied on their own terms (e.g., Adams & Swain, 2002; Connor, 1996; Cook & Bassetti, 2005a).

The richness and variety of the directions taken in recent CLI studies are not an accidental development. They stem from the greater interdisciplinarity of the research itself, which has resulted from an increasing cross-pollination among the fields of bilingualism, cognitive science, psychology, SLA, and theoretical linguistics. From the perspective of the field of bilingualism, the cross-pollination has been especially facilitated by the consolidation of the field around three journals—*International Journal of Bilingualism* (1997–), *Bilingualism: Language and Cognition* (1998–), and *International Journal of Bilingual Education and Bilingualism* (2000–)—and two regularly held conferences—the International Symposium on Bilingualism (1997, 1999, 2001, 2003, 2005, 2007), and the University of Vigo Symposium on Bilingualism (1997, 2002). These forums have brought together linguists, psycholinguists, and psychologists interested in bilingualism, language learning, and attrition, and have facilitated the exchange of ideas, theories, and methodologies across disciplinary boundaries. This exchange has also resulted in a variety of theoretical proposals, centrally or peripherally concerned with transfer.

1.6. RECENT DEVELOPMENTS IN CLI: NEW THEORETICAL ACCOUNTS

New theoretical accounts of CLI and other linguistic phenomena resulting from the interaction between two or more languages have developed against the background of general theoretical advances in the fields of linguistics, psychology, bilingualism, and second language acquisition. The first and perhaps most important theoretical development in CLI is the growing recognition of the relevance of linguistic relativity, or the Sapir-Whorf Hypothesis, to transfer research. Several years ago, Lakoff (1987) complained that with the growing dominance of generative linguistics and its cornerstone, Universal Grammar, “most ‘responsible’ scholars have steered clear of relativism. It has become a *bête noire*, identified with scholarly irresponsibility, fuzzy thinking, lack of rigor, and

even immorality” (p. 304). The current impetus for investigations into linguistic relativity has come from Lakoff’s own work, and also from the groundbreaking work of Lucy (1992a, b; 1997), Slobin (1996, 2000, 2001), and Levinson and associates (Bowerman & Levinson, 2001; Gumperz & Levinson, 1996; Levinson, 1996, 1997, 2003a, b). These scholars have pointed out that critics of linguistic relativity have traditionally oversimplified and misinterpreted Sapir’s and Whorf’s original claims about how language can influence thought (i.e., linguistic relativity), falsely assuming that Sapir and Whorf believed that language strictly determines thought (i.e., linguistic determinism). Lakoff and the other neo-relativists have also forged new, complex, and nuanced approaches to the study of the ways in which different aspects of language shape distinct modes of thought, acknowledging that some cognitive processes and modes of thought may not be affected by language at all. These developments have spurred an array of empirical studies that attest to the linguistic effects on cognitive processes and conceptual representations (Bowerman & Levinson, 2001; Gentner & Goldin-Meadow, 2003; Gumperz & Levinson, 1996; Lakoff, 1987; Levinson, 2003a; Lucy, 1992b; Niemeier & Dirven, 2000; Nuyts & Pederson, 1997; Pütz & Verspoor, 2000).

Of course, the relativistic perspective is not a prerequisite to exploring transfer, given that crosslinguistic influences are assumed within several different theoretical frameworks, including the universalist paradigm. However, the different approaches to CLI have radically different implications for cognition. Kellerman (1995, 2001) and Odlin (2002, 2003, 2005) have pointed to possible implications of the Sapir-Whorf hypothesis for transfer research, yet have not made a comprehensive attempt to link the two. The present book makes such an attempt, building on Pavlenko’s (2005a) argument that misinterpretations of Sapir’s and Whorf’s arguments also stem from the monolingual bias of the scholars who have assumed that the theory aims to describe a world where monolingual speakers differ from each other because they are locked into particular modes of thinking by their languages. This monolingual bias does not, however, come from Whorf, one of the first to champion the importance of multilingual awareness and to argue that “to restrict thinking to the patterns merely of English, and especially to those patterns which represent the acme of plainness in English, is to lose a power of thought which, once lost, can never be regained” (1956, p. 244).

Whorf’s writings clearly show his belief that additional language learning has the power to transform or enhance the speaker’s worldview. It is therefore ironic that his work was later misinterpreted as an argument for linguistic determinism, whereby the language one speaks determines one’s view of the world once and for all. Clearly, Whorf, an avid language learner who was committed to comparative linguistics, did not and could not entertain such a possibility; rather, he argued for the benefits of linguistic pluralism. Unfortunately, even though some applied linguists and bilingualism scholars remained conscious of